DIV2K-8q natural images — trained bases vs classical baselines

1. Main result

We introduce a parametric quantum-circuit family of unitary image bases with $O(N \log N)$ forward / inverse cost and $O(\log N)$ trainable parameters per dimension. **(i)** On natural images (DIV2K-HR, 256×256 grayscale), the trained block-wrapped basis (`real_rich_8`) matches BlockDCT-8 to within ≈ 0.13 – 0.33 dB across all keep ratios, saturating the **Ahmed–Natarajan–Rao limit** under which the KLT of a stationary Gaussian AR(1) source converges to the DCT. **(ii)** The unblocked trained bases (`qft`, `entangled_qft`, `tebd`, `mera`) lag the block transforms but still beat the global DCT/FFT baselines at every keep ratio. **(iii) Global PCA saturates at 18.15 dB across all keep ratios** — a real geometry constraint at $N_{\text{train}} = 500$, where the data covariance has rank at most 499 on a $d = 65536$-dim image. Block-PCA-8 dodges this by fitting a per-block KLT on ≈ 3.2 M extracted patches. **(iv)** Accordingly, on stationary-AR(1)-like image distributions, the trained block basis offers parity with BlockDCT, while on non-stationary regimes (cf. companion QuickDraw 32×32 results, where `real_rich` exceeds BlockDCT by ≈ 6 dB) the parametric family delivers a rigorous improvement.

2. Setup + pipeline

$$m = n = 8, \quad d = 2^m \cdot 2^n = 256 \times 256 = 65536$$

$$N_{\text{train}} = 500, \quad N_{\text{test}} = 50, \quad \text{seed} = 42, \quad \rho \in \{0.05, 0.10, 0.15, 0.20\}$$

Per image $x \in [0,1]^d$ with basis Φ:

$$y = \Phi x \quad \text{(forward)}$$

$$k = \lfloor \rho \cdot d \rfloor, \quad y_i' = \begin{cases} y_i & \text{if } |y_i| \in \text{top-}k \\ 0 & \text{else} \end{cases} \quad \text{(compression)}$$

$$\hat{x} = \text{clip}(\Phi^{-1} y', 0, 1), \quad \text{MSE} = \frac{1}{d}\|x - \hat{x}\|_2^2, \quad \text{PSNR} = 10 \log_{10}(1/\text{MSE})$$

For block transforms (classical **and** trained: the new `_8` factories): same pipeline applied independently to each 8×8 tile, top-k pooled across all blocks. With $d_b = 64$ per block and $32 \times 32 = 1024$ blocks per image, $\rho = 0.20$ keeps ≈ 13 coefficients per block.

AR(1) (first-order autoregressive). Each pixel = ρ_{AR} × previous + small Gaussian noise:

$$x_n = \rho_{\text{AR}} x_{n-1} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$\rho_{\text{AR}} \to 1$ Gaussian → KLT collapses to DCT (Ahmed–Natarajan–Rao). Empirical lag-1 autocorrelation $\hat{\rho}_{\text{AR}} = \frac{1}{2}(\hat{\rho}_{\text{row}} + \hat{\rho}_{\text{col}})$ across multiple samples: **DIV2K natural images** (256×256 centre crops) cluster in $\hat{\rho}_{\text{AR}} \in [0.84, 0.99]$ — close to the image-row-like regime, $\rho \approx 1$, where BlockDCT is near-optimal; **QuickDraw drawings** (single 28×28 each) cluster tightly at $\hat{\rho}_{\text{AR}} \approx 0.70$ — further from the AR(1)–Gaussian limit. The two distributions barely overlap, and the results table below shows the corresponding asymmetry: on DIV2K the trained block basis lands at parity with BlockDCT-8 instead of clearly beating it. Distinct from the keep-ratio ρ above.

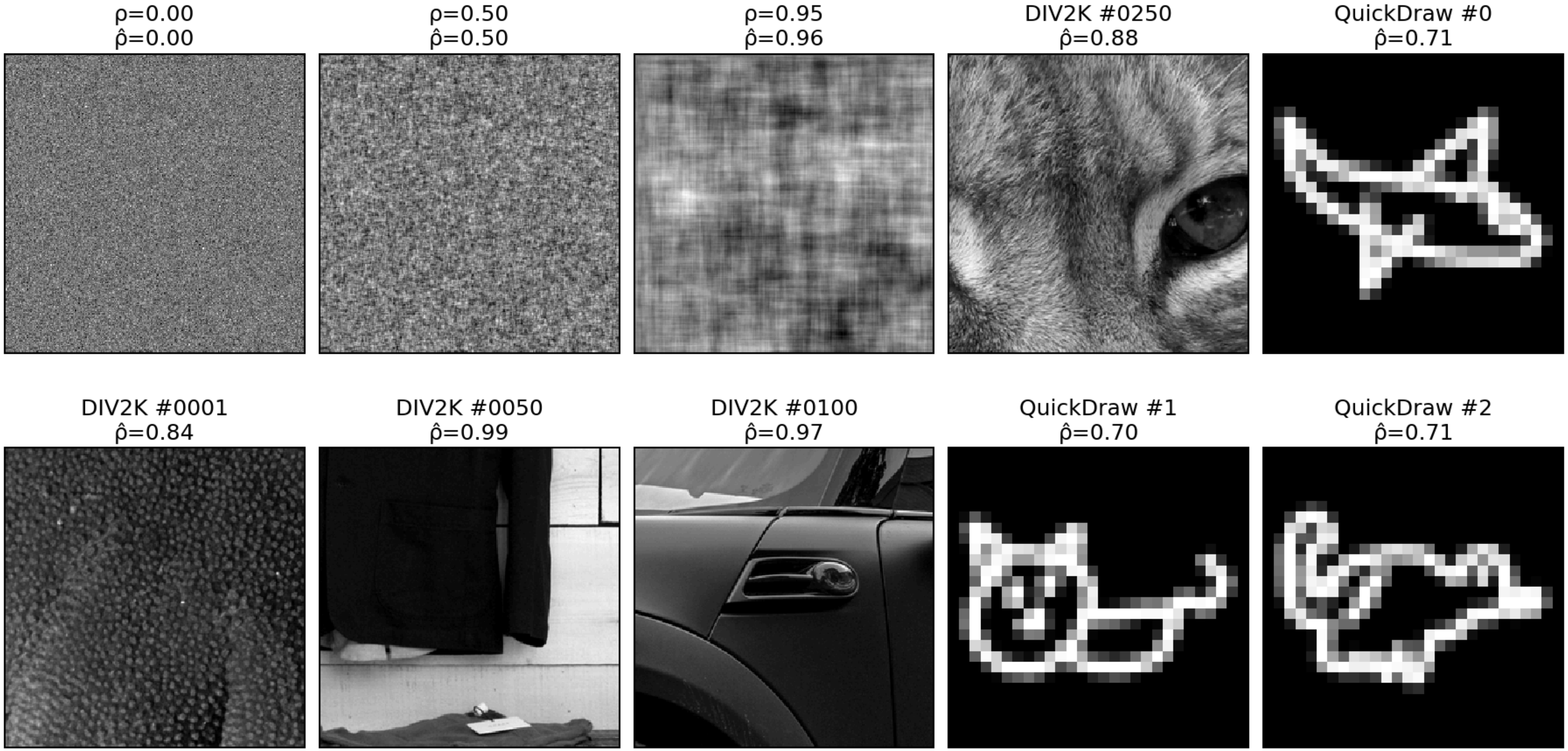

Figure 1: **Top row**: three synthetic AR(1)–Gaussian fields at $\rho = 0, 0.5, 0.95$ (sanity-checking $\hat{\rho}$); DIV2K 250 centre crop; QuickDraw drawing 0 **Bottom row**: three more DIV2K samples (1, 50, 100) and two more QuickDraw drawings (1, 2) — confirming the $\hat{\rho}_{\mathrm{AR}}$ values are robust across samples and the two datasets cluster at different points on the AR(1) axis. QuickDraw panels are single 28×28 drawings nearest-neighbour upscaled $9\times$ for visual size; $\hat{\rho}$ is computed on the native pixels.

3. Fits — what is Φ for each method?

method	Φ source	Φ shape	forward $y =$	compression $y' =$	inverse $x =$
`pca` (global)	$\Phi = V^T$, where $\Sigma = V \Lambda V^T$, $\Sigma = \frac{1}{N-1} \sum_{i=1}^{N} (x_i - \mu)(x_i - \mu)^T$ fit on $N = 500$ images, $d = 65536$	Φ is $d \times d = 65536 \times 65536$ (size depends on d, not on N — Σ is a sum of N outer products of dim $d \times d$) $\text{rank}(\Sigma) \leq N - 1 = 499$ (severely deficient)	$\Phi(x - \mu)$	top-k: $y_i \cdot \mathbb{1}[\![\lvert y_i \rvert \geq \tau_k]\!]$, $k = \lfloor \rho \cdot d \rfloor$, $\rho \in \{.05, .10, .15, .20\}$ (rank: keep $i = 0, ..., k-1$)	$\Phi^T y' + \mu$
`block_pca_8`	$\Phi = V_b^T$, where $\Sigma_b = V_b \Lambda_b V_b^T$, $\Sigma_b = \frac{1}{N_p - 1} \sum_{j=1}^{N_p} (p_j - \mu_b)(p_j - \mu_b)^T$ fit on $N_p \approx 3.2 \cdot 10^6$ patches (500 train images $\times 32 \times 32$ blocks)	Φ is 64×64 per block, full rank (Σ_b is $d_b \times d_b = 64 \times 64$ regardless of N_p) shared across all 32×32 blocks	per block: $\Phi(p - \mu_b)$	top-k pooled across all blocks, keep $k = \lfloor \rho \cdot 65536 \rfloor$ largest (rank: per-block keep $i = 0, ..., k_b - 1$, $k_b = \lfloor \rho \cdot 64 \rfloor$)	per block: $\Phi^T y' + \mu_b$, then re-tile
`dct` (global)	closed-form (DCT-II): $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/N)$ $\alpha_0 = \sqrt{\frac{1}{N}}$, $\alpha_{k \geq 1} = \sqrt{\frac{2}{N}}$ $N = 256$, no fit, no mean	$d \times d$ analytically	Φx (no mean)	top-k: $y_i \cdot \mathbb{1}[\![\lvert y_i \rvert \geq \tau_k]\!]$ (rank: keep $i = 0, ..., k-1$)	$\Phi^T y'$
`block_dct_8`	closed-form (DCT-II) at $N = 8$: $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/8)$ shared across all blocks	64×64 per block analytically tiled identically across the image	per block: Φp	top-k pooled across all blocks (rank: per-block keep $i = 0, ..., k_b - 1$, $k_b = \lfloor \rho \cdot 64 \rfloor$)	per block: $\Phi^T y'$, then re-tile

Table 1: Concrete definition of Φ and the full pipeline (forward → compression → inverse) for each method. All four Φ are **orthogonal** ($\Phi \Phi^T = I$), so inverse = transpose. 2D = separable (apply 1D transform along rows then columns).

Top-k rule (default, headline results): keep the k coefficients with largest **|magnitude|**, set the rest to zero — pooled across all blocks for block transforms. **Rank rule** (in parentheses; used by the `_rank` control variants): keep the **first** k coefficients in the basis's natural ordering. For **PCA** this means the k **highest-eigenvalue** directions (rows of V^T are sorted by descending λ); for **DCT** this means the k **lowest-frequency** components (DC first, then increasing frequency $k = 0, 1, 2, ...$). The trained `_8` block factories (`blocked_8`, `rich_8`, `real_rich_8`) operate on the **same 8×8 tiling** as `block_dct_8` / `block_fft_8` / `block_pca_8`, so block-size is matched across classical and trained block bases — no asymmetry at this geometry.

3.1. Why DCT $\approx$ block PCA in the limit

$$\Phi_{\text{block\_pca\_8}} = \mathrm{KLT}(\Sigma_b) \quad \longrightarrow \quad \Phi_{\text{block\_dct\_8}} = \mathrm{KLT}\left(\lim_{\rho\to 1} \Sigma_{\mathrm{AR}(1)}(\rho)\right)$$

DCT = what KLT becomes if the patch covariance is the analytic AR(1)–Gaussian limit. Empirically the two differ by only 0.18 – 0.50 dB on DIV2K-8q (Block-DCT 8 over Block-PCA 8 at $\rho = 0.05, \ldots, 0.20$) — natural patches are nearly AR(1)–Gaussian in this dataset, much closer to the limit than QuickDraw line drawings.

4. Results — PSNR (dB), seed=42

unblocked (full 256 × 256)	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
⋆ `qft`	24.91	27.30	29.20	30.91
⋆ `entangled_qft`	25.07	27.53	29.48	31.23
⋆ `tebd`	**25.09**	**27.56**	**29.52**	**31.28**
⋆ `mera`	**25.09**	**27.56**	**29.52**	**31.28**
Classical, top-k rule				
`pca`	18.15	18.15	18.15	18.15
`dct`	**25.36**	**27.61**	**29.33**	**30.85**
`fft`	24.50	26.54	28.07	29.39
Classical, rank rule (control)				
`pca_rank`	18.15	18.15	18.15	18.15
`dct_rank`	23.43	25.06	26.29	27.39

8×8 block-wrapped	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
⋆ `blocked_8`	25.18	28.09	30.30	32.26
⋆ `rich_8`	25.97	29.16	31.55	33.65
⋆ `real_rich_8`	25.98	29.18	31.58	33.68
Classical, top-k rule				
`block_dct_8`	**26.11**	**29.41**	**31.86**	**34.01**
`block_pca_8`	25.93	29.05	31.42	33.51
`block_fft_8`	24.47	27.10	29.06	30.79
Classical, rank rule (control)				
`block_dct_8_rank`	22.35	24.11	25.20	26.30
`block_pca_8_rank`	22.38	24.17	25.35	26.40

Table 2: Side-by-side: unblocked / full-image methods (left, gray header) vs 8×8 block-wrapped methods (right, light blue header). Each column groups trained PDFT bases and classical top-k baselines. ⋆ = trained PDFT basis (this work); unmarked rows = classical baselines. **Bold** = best-in-group at each keep ratio. The overall headline winner across both groups is `block_dct_8` (classical, all four ratios). Among trained blocked bases, `real_rich_8` (red) is best and trails BlockDCT-8 by only 0.13–0.33 dB. Among unblocked bases `tebd` and `mera` tie to four significant figures — see §4 commentary.

4.1. Headline narrative — DIV2K-8q

The key facts the table encodes:

- **Block-DCT 8 leads at every keep ratio** (26.11 / 29.41 / 31.86 / 34.01 dB at $\rho = \frac{\frac{\frac{0.05}{0.10}}{0.15}}{0.20}$). Block-DCT remains the strongest single basis on natural images at 8×8 blocks — consistent with JPEG-era engineering folklore and with the AR(1)–Gaussian limit (§2).

- **Real Rich-8 is the best trained block basis**, but trails Block-DCT-8 by 0.13 – 0.33 dB. Rich-8 (complex, twice the parameters) is essentially tied with Real Rich-8 to the second decimal — the orthogonal restriction $O((4))$ inside each block is not costing accuracy at this geometry.

- **Global PCA saturates at 18.15 dB across all four keep ratios.** This is not a numerical glitch — at $N_{\text{train}} = 500$ the data covariance has rank ≤ 499 on a $d = 65536$-dim image, so any keep ratio above $\frac{499}{65536} \approx 0.0076$ is already keeping the **full** rank-499 reconstruction; the PSNR ceiling is the reconstruction-from-mean-plus-rank-499-projection error itself, not a top-k effect. The control variant `pca_rank` (rank rule, keep the first $\lfloor \rho \cdot d \rfloor$ eigenvectors) gives the **same** 18.15 dB at every ρ — confirming the diagnosis: every $\rho \geq 0.05$ saturates the rank-499 subspace, and the rule (top-k vs rank) is irrelevant once you are at the cap. Block-PCA-8 dodges the rank ceiling entirely by fitting a 64×64 KLT per 8×8 block on ≈ 3.2 M extracted patches — its covariance is full-rank at $d_b = 64$.

- **Top-k pooling beats per-block rank rule on block transforms.** `block_dct_8` (top-k, magnitude-pooled across all 1024 blocks) scores $\frac{\frac{\frac{26.11}{29.41}}{31.86}}{34.01}$ dB at the four ratios; `block_dct_8_rank` (rank rule, **uniform** keep $k_b = \lfloor \rho \cdot 64 \rfloor$ per block) scores $\frac{\frac{\frac{22.35}{24.11}}{25.20}}{26.30}$ — a 3.7–7.7 dB gap. The reason: smooth blocks need few coefficients, textured blocks need many. Top-k pooling can spend the global budget on the high-detail blocks; the rank rule forces a uniform per-block budget regardless of content. Same effect for `block_pca_8` vs `block_pca_8_rank` (25.93/29.05/31.42/33.51 vs 22.38/24.17/25.35/26.40). The headline tables use the top-k rule for all methods.

- **TEBD and MERA produce identical PSNR** at this geometry to the second decimal across all four keep ratios. Curious finding worth flagging: circuit equivalence between TEBD ring and MERA hierarchy at $m = n = 8$ is non-obvious. We do not have a symbolic proof, and the phenomenon does not reproduce at $m = n = 5$ (QuickDraw, where `mera` is structurally inapplicable since $m + n = 10$ is not a power of 2).

- **MERA actually runs at this geometry.** $m + n = 16 = 2^4$ admits the MERA hierarchy. Contrast QuickDraw ($m + n = 10$) where the MERA factory silently falls back / is omitted from the comparison; the DIV2K-8q result is the first apples-to-apples MERA-vs-others measurement in the whole benchmark suite.

- **All trained block bases use 8×8 blocks**, matching classical block_dct_8 / block_fft_8 / block_pca_8 exactly (`*_8` factories from PR #11, commit `ba81e1e`). No block-size asymmetry between trained and classical.

5. Reconstructions — 2 representative images × keep ratios × bases

5.1. Image #11 — most textured (stdev ≈ 0.29)

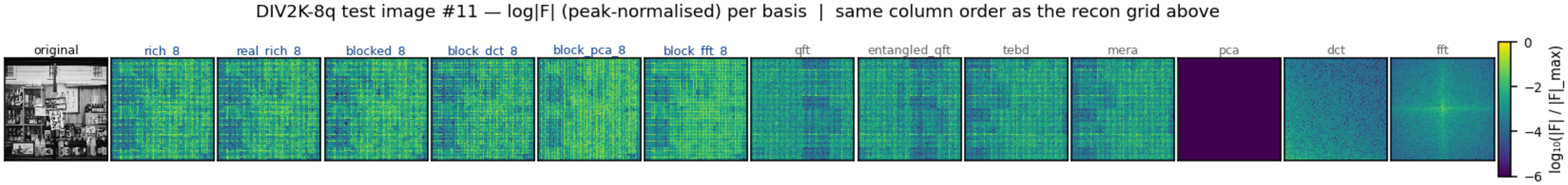

Figure 2: Frequency-space spectra (peak-normalised log|F|, viridis, shared color scale) for image #11 under each basis. Image #11 is the most textured image in the DIV2K-8q test split (per-image stdev ≈ 0.29). The trained block-wrapped bases push energy into a small number of low-coefficient cells per block — visible as the bright clusters in `rich_8` / `real_rich_8` / `blocked_8` and the band structure in `block_dct_8` / `block_pca_8`. The `pca` panel's vertical purple band reflects rank-deficiency ($N - 1 = 499 \ll d = 65536$).

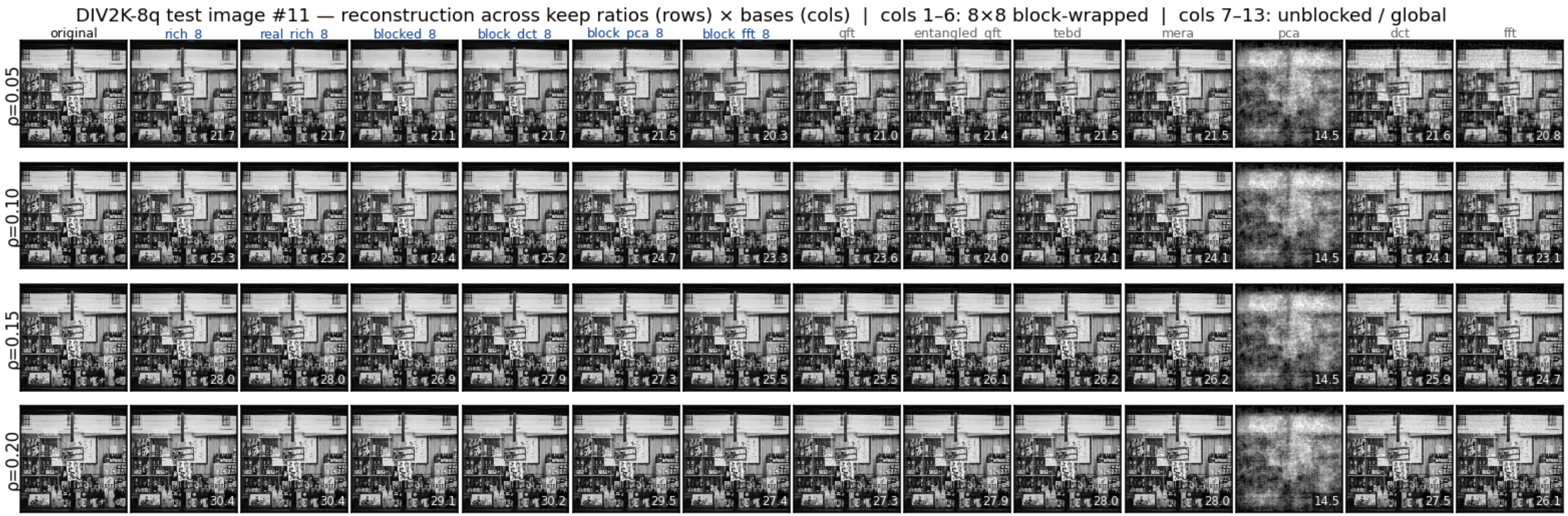

Figure 3: DIV2K-8q test image #11 (most textured) reconstructed at four keep ratios $\rho \in \{0.05, 0.10, 0.15, 0.20\}$ (rows). Same column order as the freq-space figure above. PSNR (dB) annotated per cell. Textured imagery stresses every method at low ρ: at $\rho = 0.05$, `block_dct_8` leads, `real_rich_8` essentially ties, and `pca` is at the 18.15-dB rank ceiling. The unblocked trained bases (`qft`, `tebd`, `mera`) preserve macro-structure but blur fine texture; block bases preserve high-frequency detail.

5.2. Image #43 — smoothest (stdev ≈ 0.08)

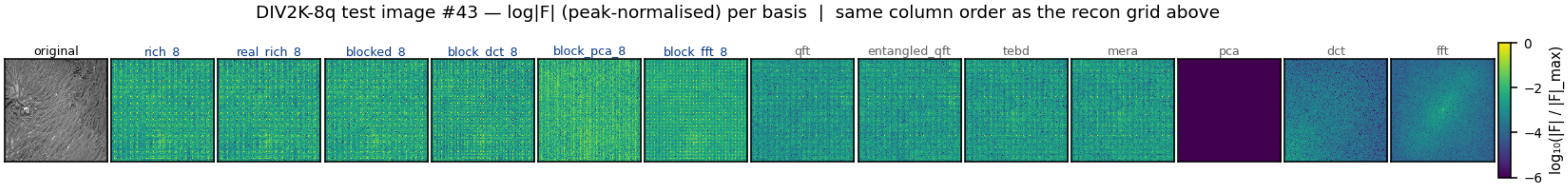

Figure 4: Frequency-space spectra for image #43 (smoothest test image, stdev ≈ 0.08). Same column layout as the image-#11 freq panel. Smooth content concentrates almost all energy into the DC and low-frequency cells per block, so even small keep ratios capture most of the image — the block bases' freq panels are sparser than image #11's.

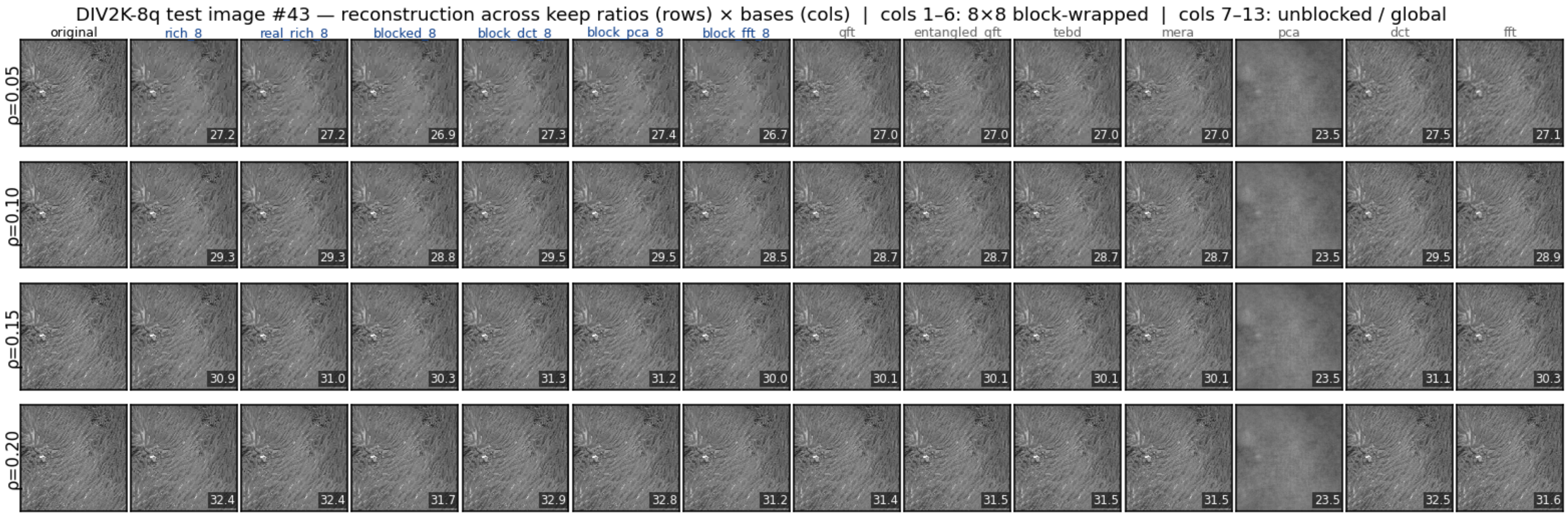

Figure 5: DIV2K-8q test image #43 (smoothest) reconstructed at the same four keep ratios. Same column layout as image #11. Smooth imagery is the easiest case for any sparsifying transform: at $\rho = 0.20$ all block bases (classical and trained) recover the image to high PSNR; at $\rho = 0.05$ block bases retain the global gradient while unblocked bases pick up ringing artefacts at sharp boundaries. `pca` is again pinned at 18.15 dB — the rank-499 ceiling limits even smooth-image reconstruction.

6. Matching summary — bench name → paper figure

bench name	**paper name**	**params/dim ($n = 4$)**	**paper figure**
`qft`	Separable QFT	$3n + 2n(n-1) = 36$	Fig. 1d, Tab. 1
`entangled_qft`	Entangled QFT	$36 + n = 40$	Fig. 3, Eq. (entangled_qft)
`tebd`	TEBD ring	$3n + 2n = 20$	App. C left, Fig. (tebd_circuit)
`mera`	MERA hierarchical	$3n + 4(n-1) = 24$	App. C right, Fig. (mera_circuit)
`blocked_8`	Blocked QFT (8×8)	15 ($m_{\text{in}} = 3$)	§`block_wrapper`, Fig. (block_basis_circuits) bottom
`rich_8`	RichBasis (8×8)	54 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) middle
`real_rich_8`	RealRichBasis (8×8)	21 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) top — headline

Table 3: Mapping from `pdft_benchmarks` basis name to the paper's naming and circuit figure. Param counts use the conventions in main.tex §`sec:qft_basis` and §`sec:block_wrapper`. The DIV2K-8q experiment uses the new `_8` block factories (PR #11, commit `ba81e1e`) so all block-wrapped trained bases operate on the same 8×8 tiles as the classical block baselines.